

EXPERIENCE

Lawrence Berkeley National Laboratory - *Lab Affiliate*, Berkeley, CA August 2025 - present

Principal Investigator: Bethany Goldblum - *UC Berkeley*

- Doctoral research.

Lawrence Livermore National Laboratory - *Graduate Intern*, Livermore, CA June 2025 - present

Faculty mentor: Jesse Norris - *Nuclear Criticality Safety Division*

- Developed CRISP (CRITICAL Simulation Pipeline) for parallelizing a Monte Carlo simulation code, COG. Utilized world class HPC machines Dane and Ruby with Slurm workload manager for parallelizing the 3500 ICSBEP reactor benchmarks. Used modern Python software development techniques such as Poetry dependency management, Git version control with YAML deployment pipeline, SQL database management, Jinja2 templating, and object-oriented programming.

Brookhaven National Laboratory - *Nuclear Science Intern*, Upton, NY 2022 - 2024

Faculty mentor: Gustavo Nobre - *National Nuclear Data Center*

- Conducted nuclear data R&D, applying machine learning, random matrix theory, and Bayesian inference to improve heavy-nuclei resonance classification. Developed synthetic training datasets using Porter–Thomas distributed capture widths and implemented iterative learning workflows for enhanced model accuracy. Performed Monte Carlo neutron transport simulations and perturbative sensitivity analyses of thermal $1/v$ neutron cross sections for critical reactor benchmark.

Truman State University - *Student Astronomy Researcher*, Kirksville, MO 2021 - 2022

Faculty mentor: Vayujeet Gokhale - Department of Physics

- Calculated trajectories of Starlink satellites to optimize telescope viewing plans. Researched long-exposure luminosity data corruption due to Starlink satellite interference. Developed a GUI for Truman astronomy students. Wrote and submitted a proposal for the TruScholars grant.